

Nonlinear Systems I: Phase-Plane Analysis

Understanding a second-order nonlinear system from its trajectories in the state plane.

Transfer functions and poles describe only **linear** systems. A second-order nonlinear system can still be understood graphically through its **phase portrait**: the family of trajectories drawn in the $(x_1, x_2) = (x, \dot{x})$ plane. In this tutorial you will:

- draw the **vector field** and trajectories of a damped pendulum,
- locate and classify its equilibria (stable focus vs. saddle), and
- see a **limit cycle** in the Van der Pol oscillator – a phenomenon with no linear analog.

Uses `ode45` (no toolbox required). Run with `publish('PhasePlane.m')`.

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The pendulum as a state-space system

$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$ with $x_1 = \theta$, $x_2 = \dot{\theta}$ becomes

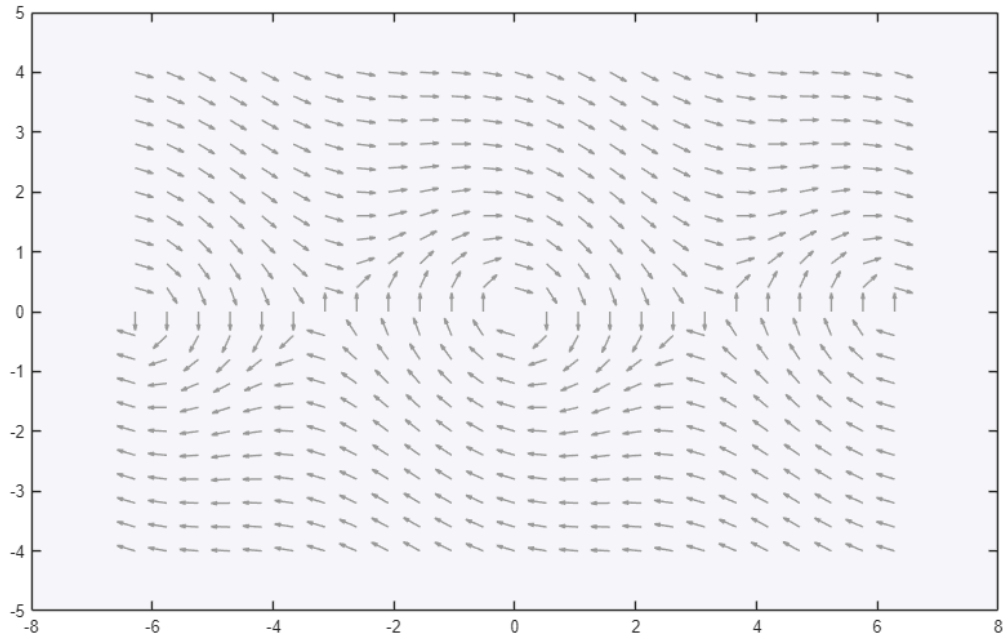
$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -bx_2 - \sin x_1.$$

```
b = 0.3;
pend = @(t,x) [x(2); -b*x(2) - sin(x(1))];
```

The vector field

At every point the arrow $(\dot{x}_1, \dot{x}_2)$ shows which way a trajectory moves. Arrows are normalized to unit length so direction is readable everywhere.

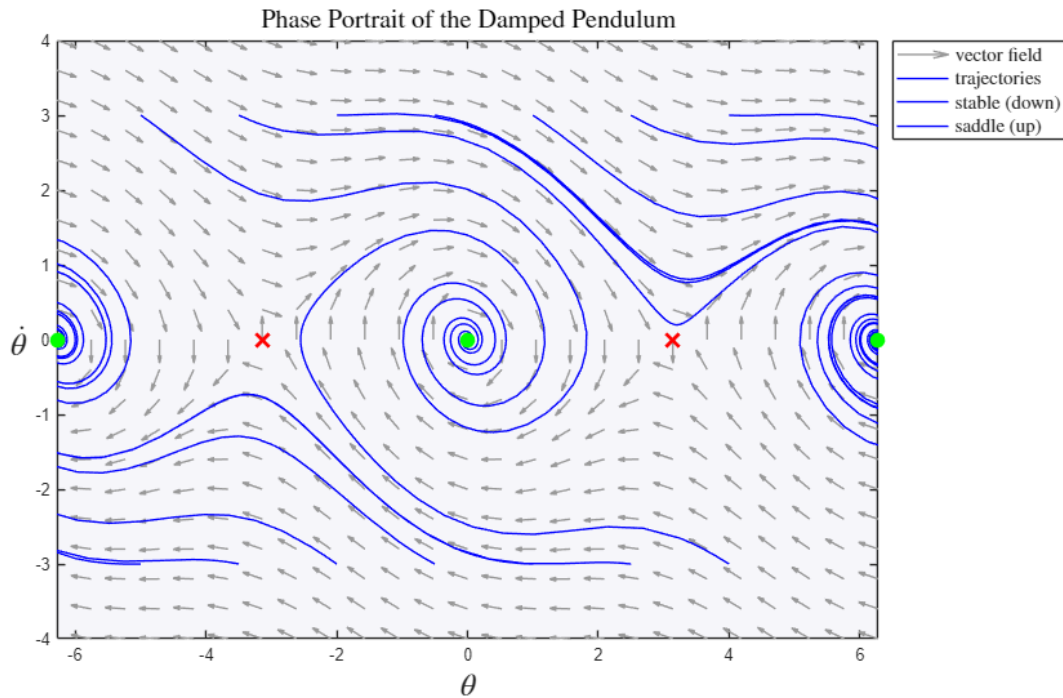
```
[X1,X2] = meshgrid(linspace(-2*pi,2*pi,25), linspace(-4,4,21));
U = X2;
V = -b*X2 - sin(X1);
Nrm = hypot(U,V); Nrm(Nrm==0) = 1;
figure
quiver(X1,X2, U./Nrm, V./Nrm, 0.5, 'Color',[.6 .6 .6])
hold on
```



Trajectories and equilibria

Overlay trajectories from several initial conditions. Equilibria sit at $(k\pi, 0)$: the **down** positions $(0, \pm 2\pi)$ are stable foci (the damping spirals trajectories in), while the **up** positions $(\pm\pi)$ are saddles (unstable).

```
for x10 = -5:1.5:5
    for x20 = [-3 3]
        [~,xx] = ode45(pend,[0 25],[x10;x20]);
        plot(xx(:,1),xx(:,2),'b')
    end
end
plot([-2*pi 0 2*pi],[0 0 0],'go','MarkerFaceColor','g','MarkerSize',8)
plot([-pi pi],[0 0],'rx','MarkerSize',11,'LineWidth',2)
hold off
axis([-2*pi 2*pi -4 4])
legend('vector field','trajectories','stable (down)','saddle (up)', ...
    'Interpreter','latex','FontSize',11,'Location','northeastoutside')
title('Phase Portrait of the Damped Pendulum','Interpreter','latex','FontSize',15)
ylabel('$\dot{\theta}$','Interpreter','latex','FontSize',18); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$\theta$','Interpreter','latex','FontSize',18)
```



Classifying equilibria with the Jacobian

Linearizing about an equilibrium x^* gives the local behavior. The Jacobian is $J = \begin{bmatrix} 0 & 1 \\ -\cos \theta & -b \end{bmatrix}$.

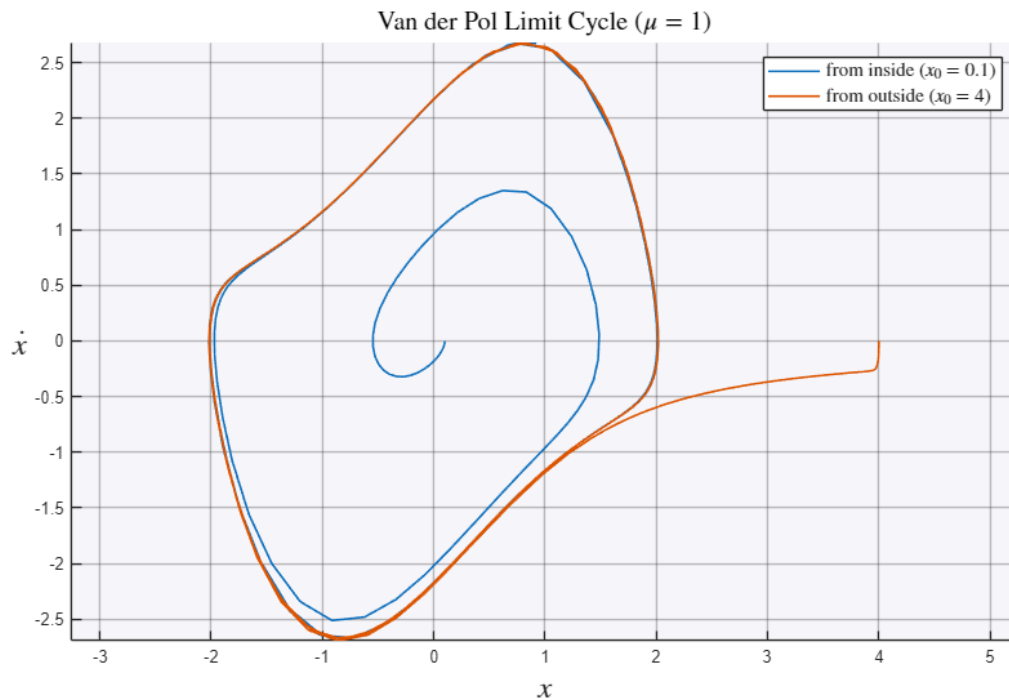
```
J = @(th) [0 1; -cos(th) -b];
fprintf('Down (theta=0): eigenvalues %s -> stable focus\n', mat2str(eig(J(0)).',3))
fprintf('Up (theta=pi): eigenvalues %s -> saddle (unstable)\n', mat2str(eig(J(pi)).',3))
```

```
Down (theta=0): eigenvalues [-0.15+0.989i -0.15-0.989i] -> stable focus
Up (theta=pi): eigenvalues [-1.16 0.861] -> saddle (unstable)
```

A limit cycle: the Van der Pol oscillator

$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$. For $\mu > 0$ **every** trajectory – started inside or outside – is drawn onto a single closed orbit, the **limit cycle**. A linear system can only spiral in or out; a sustained, amplitude-stable oscillation like this is intrinsically nonlinear.

```
mu = 1;
vdp = @(t,x) [x(2); mu*(1-x(1)^2)*x(2) - x(1)];
figure
hold on
for r0 = [0.1 4]
    [~,xx] = ode45(vdp,[0 25],[r0;0]);
    plot(xx(:,1),xx(:,2),'LineWidth',1.2)
end
hold off
grid on; axis equal
legend('from inside ($x_0=0.1$)', 'from outside ($x_0=4$)', ...
    'Interpreter','latex','FontSize',11)
title('Van der Pol Limit Cycle ($\mu=1$)', 'Interpreter','latex','FontSize',15)
ylabel('$\dot{x}$', 'Interpreter','latex','FontSize',18); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$x$', 'Interpreter','latex','FontSize',18)
```



Summary

- A phase portrait shows the global behavior of a 2nd-order nonlinear system at a glance: where it settles, and where it cannot.
- The Jacobian classifies each equilibrium locally (focus, node, saddle).
- Nonlinear systems support behaviors – multiple equilibria, limit cycles – that linear systems simply cannot.

Next: `LyapunovStability.m` certifies stability **without** solving the equations.

Try it yourself

- Raise the damping b and notice the pendulum trajectories spiral into the down equilibrium faster (a tighter focus).
- Set the Van der Pol μ to 3 and watch the limit cycle sharpen into a relaxation oscillation.